UNIVERSITY OF MORATUWA

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BSc Engineering Honours Degree

Semester 4 Examination: 2024

CS3023: SOFTWARE ENGINEERING

- **Time allowed: 2 Hours**
- **ADDITIONAL MATERIAL: None**
- **August 2024**

INSTRUCTIONS TO CANDIDATES:

1. This paper consists of 3 questions in 4 pages.
2. Answer all 3 questions.
3. Start answering each of the 3 main questions on a new page.
4. The maximum attainable mark for each question is given in brackets.
5. This examination accounts for 60% of the module assessment.
6. This is a close-book examination. **NB:** It is an offence to be in possession of unauthorised material during the examination.
7. Only calculators approved and labelled by the Faculty of Engineering are permitted.
8. Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions made on the script.
9. In case of any doubt as to the interpretation of the wording of a question, make suitable assumptions and clearly state them on the script.
10. This paper should be answered only in English.

Question 1 (30 marks)

(i) Critically examine the future of software engineering methods and techniques considering the generative AI tools used in the software lifecycle. [5 marks]

(ii) Compare and contrast the software development under a proper software process and without using such a software process. To support your answer you may use an appropriate example software process model. [5 marks]

(iii) Models with abstraction provide useful analysis options to provide a focused view of a selected problem. Also, the models with abstraction provide useful design considerations to provide better solutions. Yet models are not completely accurate. Discuss this observation with respect to the major activities of a software lifecycle while providing relevant models for an example project of your choice. [10 marks]

(iv) Suggest a suitable high-level architecture model to combine xUML and Generative AI technologies to produce fully automated software creation without devteam active involvement. [4 marks]

(v) Describe the following:

- a. Volatile requirements
- b. CoCoMo Post Architecture model
- c. Version Control $[3 \times 2 \text{ marks}]$

Question 2 (40 marks)

Department of Immigration and Emigration (DIE), Sri Lanka has started a software system development project that aims to automate the processes of passport issuance for Sri Lankan citizens. Once completed passport-seeking citizens can complete an online application, upload necessary documents, pay the application fee online, and reserve an appointment with the nearest regional branch or authorised agents to provide biometrics (photo, finger scans and personal verification). These collected data will be processed using a risk modelling module and grouped into high-risk (Red), medium-risk (Yellow), and low-risk (Green) channels of processing, with differential authority levels for approval. Once completed, the issuance of passports will be carried out at the head office and will be delivered to the given address. You have been recruited as a lead software engineer for the project by the software development company that develops the solution.

(i) Identify five key stakeholders of this system. Briefly justify your answer stating their main goal/objective with respect to the system. [5 marks]

(ii) Draw five UML use case diagrams for each identified stakeholder. Text descriptions are not required. [5 marks]

(iii) State and describe three (3) functional and three (3) non-functional requirements for the system while identifying necessary metrics to measure each. [6 marks]

(iv) Draw a suitable UML sequence diagram for the bio-metric capturing process. State your assumptions. [4 marks]

(v) Briefly describe the Architecture at large of the proposed system using suitable illustrations. [5 marks]

(vi) Prepare complete architecture diagrams using the following perspectives for the proposed system with sufficient finer-level details.

- a. Software component level breakdown using a class diagram [5 marks]
- b. Software component level interactions using interaction diagrams [6 marks]
- c. High-level structure using a package diagram [4 marks]

Question 3 (30 marks)

a) Explain how container-based virtualization is different from Virtual Machine (VM) based virtualization. You may use diagrams to assist your answer. [5 marks]

b) The National Building Research Organization (NBRO) has contracted your company to develop LandWatchLK a software which will assist their in-house scientists to analyse Sri Lanka using satellite imagery. The application will periodically (i.e. daily) download raw satellite imagery from multiple sources over the Internet, carry out some pre-processing, and store them locally on the server. Scientists specify various scientific analyses by selecting source images (location, dates, resolution, type) and operations to carry out on them. Depending on the size/number of source images and the complexity of operations involved, an analysis can take minutes or hours. Scientists interact with the system through a Web-browser based interface. Since storage can run into terabytes of disk space there is a need to check for old intermediate and result files and delete them. The project is being funded by a large one-off grant from the World Bank. NBRO directors are undecided about whether to host LandWatchLK in-house or on a public cloud server. Prepare a memo to the director board making your recommendation. You should highlight the pros and cons of both approaches. Your answer must be relevant to LandWatchLK. Assumptions made must be clearly stated. (Irrelevant/generic answers will be given zero marks) [6 marks]

c) Piyal is working as a Software Engineer for a large software company. Recently, one of his code changes "broke the build" of the project he was working on.

- i. Explain briefly what it means to "break the build" and how in a DevOps environment, such a problem can be flagged. [3 marks]
- ii. Suggest an approach which Piyal can use to avoid making such mistakes in future. [2 marks]

d) Briefly explain what Infrastructure as Code is and the problem it attempts to solve. [4 marks]

e) Explain why object-oriented programming languages such as Java and C# are not suited for developing real-time systems. [3 marks]

f) Explain why an Android Operating System (OS) powered computing platform is not suitable for developing real-time systems. Suggest at least one (1) OS suitable for such systems and identify the basic components such an OS would have. [7 marks]

END OF THE PAPER